Choosing a Classification Method for Measuring Government Criticism in Armenian Civil Society Texts

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Abstract

How should researchers measure criticism in low-resource language texts? This paper aims to address this question by comparing bag-of-words and contextual BERT embedding representations paired with multiple classifiers. More specifically, I identify the strenghts and limitaitons of these approaches by looking at their performance when classifying a hand-coded training set of 642 documents from five Armenian watchdog civil society organizations as critical or non-critical of the government. Methodologically the results reveal two important findings. I find that the choice of text representation matters substantially more than the choice of classifier as BERT embeddings improve accuracy by 4 percentage points over bag-of-words, while switching classifiers within the embedding space does not make much difference. Second, fine-tuning BERT's transformer layers underperforms the simpler frozen-embedding approach (59.2% vs. 72.1% accuracy), showing that with only 642 labeled documents, adding model flexibility hurts rather than helps. Substantially, applying the best classifier to the full corpus of five Civil Society Organizations publications spanning 2014 to 2024,reveeals divergenccies in the critical stance of CSO towrads teh govement.

1 Introduction

"I am the government… no one can have a position that contradicts my position." Prime Minister Pashinyan made this statement in August 2025, justifying the dismissal of a senior official who had urged the government to comply with the Stockholm Arbitration Court's ruling and halt the nationalization of the Electric Networks of Armenia (Grigoryan, 2025). This dismissal was not an isolated incident. The arbitrary detention of clergy, the use of police and security services against the opposition, and the ruling party's use of administrative leverage to influence local elections all signal a clear backsliding. In this context, think tanks, opposition figures, and experts point to a puzzle: Armenian "watchdog" CSOs, which played a central role in challenging authoritarian rule during the 2018 Velvet Revolution, have remained largely silent in response to Pashinyan's own authoritarian practices (RCDS, 2024). These assessments also resonate well with the V-Dem Civil Society Participation Index, where Armenia's score declines from 0.89 in 2019 to 0.67 in 2024 (V-Dem, 2025). Armenia's NGO sector is heavily dependent on external funding and weakly embedded in society, which constrains its responsiveness and legitimacy (Ishkanian, 2008, 2014; Paturyan, 2014). Comparative research from Georgia further suggests that such disconnection facilitates civil society demobilization during democratic backsliding by pro-Western incumbents, as donor support often persists despite rising authoritarianism (Stefes & P1). This raises an important question: have Armenian watchdog CSOs actually reduced their criticism of the government since the revolution? And if so, why? There are several competing explanations. One hypothesis is donor-driven: the current government, while increasingly authoritarian, is pro-Western, and major funders such as USAID and the EU may be reluctant to finance criticism of a geopolitical ally. Another is organizational: brain drain following the revolution depleted civil society of its most active members, many of whom joined the government. Last but not least, it is equally possible that CSOs have not actually gone silent but have shifted to a more tacit modes of advocacy. While each of the hypothesis identified has strong theoretical basis, testing any of them is fundamentaly a text classification problem which requires a reliable measure of CSO criticism. This paper addresses the measurement side of that question. I use predictive machine learning tools to construct a measurement instrument that serves a descriptive inference goal (Breiman 2001). More specifically I

compare bag-of-words and contextual BERT embedding representations paired with Naive Bayes, Logistic Regression, Random Forest, Gradient Boosting, and a Stacking Enseme to identify the best approach for classifying approximately 4,100 publications spaning from 2014 till 2024, from five Armenian watchdog CSOs as critical or non-critical of the Armenian government. Using a hand-coded training set of 642 documents from the identified 5 CSOS, I find that the choice of text representation matters substantially more than the choice of classifier. BERT embeddings improve accuracy by 4 percentage points over bag-of-words, while the best classifier improvement within the embedding space is approximately 1 percentage point. All BERT-based classifiers produce very similar results across the entire corpus. This essentially indicates that inferences about CSO behavior are robust to classifier choice when using contextual representations. As a preliminary application, I classify the full corpus and estimate a document-level logistic regression with organization fixed effects to formally test how did the stance of five watchdog CSOs in Armenia evolve after 2018. To this end the paper is thematically organized into two parts. First, I develop and validate a text classification measure by systematically comparing text representations and classifiers. Then, I apply this measure to the full corpus to assess whether and how CSO criticism changed after 2018.

2 Background, Data, and Methods

2.1 The challenge of measuring criticism

Identifying critical positions in texts poses a number of methodological challenges that can significantly impact measurement accuracy. First, it is often difficult to accuratley identify the target of criticism. In this context, CSO publications often target not only their national governments government, but also other states, international and supranatural organizations and finally other non-profit organizations. Consequently, a document containing a critical tone could convey anything from disapproval of Russia's foreign policy to criticism of the EU visa liberalization process. This problem was particularly evident during the training data preparation, as approximately 18% of documents contained criticism of foreign entities, and not the Armenia government. Therefore, a classifier that captures "critical tone" without distinguishing the target of criticism will systematically misclassify these documents. Furthermore, it is important to acknowedlege that in non-democractic regimes, protests and expressions of criticism directed at the goverment differ significantly from those found in democratic regimes. Given the possiblity of state-sanctioned repression, citizens and organizations avoid voicing criticism in overtly explicit and negative terms, instead using more implicit formulations or even legal references. In China, for example, citizens often rely on 'rightful resistance' to legitimize their grievances against local authoritie by referencing CCP's own ideology and symbols (O'Brien, 1996). Consequently, identifying this kind of criticism requires attention to the wording and tone, not just the vocabulary.

Another interrelated problem arises from the fact that critical and non-critical documents often use similar vocabulary. Civil society organizations publish reports on elections, corruption, transparency, and human rights, regardless of whether they are critical of the government or simply reporting facts. Ultimatley, words like "corruption," "elections," and "accountability" appear in both critical and non-critical documents. The "bag of words" model treats them equally regardless of context, but the phrase "the government failed to address corruption" is critical, while "a civil society organization published a report on corruption trends" is not (Grimmer, Roberts, and Stewart

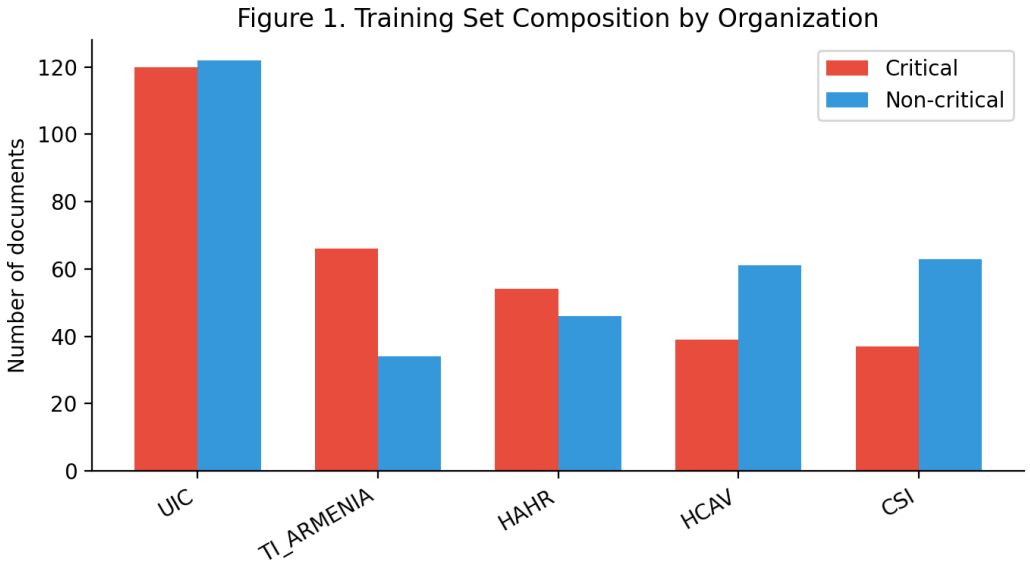
2022).

Taken together, these problems correspond to the specific capabilities of various machine learning methods. It therefore becomes a supervised classification problem: the response variable is a binary label indicating criticism or lack thereof, and the predictors are textual features. A bag-of-words model with a naive Bayes classifier processes each word independently and is unable to distinguish the object of criticism. Contextual embeddings from a pre-trained transformer like BERT encode the surrounding context, so "the government failed" and "Russia interfered" create different vector representations even though they use the same vocabulary.Within the embedding space, tree-based classifiers can capture nonlinear patterns that a linear model might miss.

2.2 Data

The corpus is built on the publications of five prominent Armenian watchdog civil society organizations: the Union of Informed Citizens, the Helsinki Citizens' Assembly Vanadzor, Transparency International Armenia, Civil Society Institute and lastly Helsinki Association for Human Rights.The selection of these organizations was guided specifically by the theoretical puzzle. Hence three selection criterias were used. First these CSOs were among the most active civil society actors during the 2018 revolution, with direct participation in the protests. Second, they hold a clear mandate to exercise public oversight and are recognized by international organizations. And finally, they maintain publicly accessible archives of their publications covering the period from 2014 to 2024. I collected the data by scraping each organization's website using the httr and rvest packages in R. The full corpus contains approximately 4,100 documents with usable text content: UIC (1,538), CSI (485), HCAV (1,559), HAHR (218), and TI Armenia (294).

I hand-coded a training set of 642 documents sampled across all five organizations. Each document was coded for two binary variables: whether it is critical of the Armenian government and whether it criticizes foreign actors. A document is coded as critical if it explicitly or implicitly criticizes the actions, policies, or inaction of the Armenian government or its officials. Across the training set, 316 documents (49.2%) are coded as critical. To ensure target-oritented analysis, before the mian classification process I have filtered out critical publications that target foreign entities to ensure that the assigned label reflects criticism directed specifically to the Armenian government. As a consistency check, I had ChatGPT independently code a subset of documents using the same coding rules, achieving 89.6% agreement. This is not a substitute for formal intercoder reliability with a second human coder, but it serves as a useful baseline for identifying systematic labeling inconsistencies, particularly for documents involving implicit criticism. Consistent with the framework proposed by Egami et al. (2024) this triangulation provides a cross-check on label reliability.



2.3 Methods

Following Barbera et al. (2021), I comprehensivly assess the models by dividing the analysis into two stages. The first part focuses on text representation, examining the transformation of the source text into numerical features. The second part evaluates classifiers, and explores how the algorithms employed predict labels based on these numbers. This seperation, in turn, helps determine whether improvements in accuracy must be attributable to richer input data or to more flexible models.

Text representations. I consider three ways of representing each document: (a) bag of words, (b) frozen BERT embeddings (c) fine-tuned BERT.

The first representation is *bag-of-words*: after tokenization, the removal of punctuation, symbols, and numbers, and the exclusion of Armenian stopwords using a custom list,[1] each document becomes a vector of word counts in a document-feature matrix. This representation is simple and transparent, but it treats each word independently regardless of context, effectively tallying vocabulary without comprehending the actual meaning of the sentence.

The second representation is *frozen BERT embeddings*. Instead of training a text classifier from scratch, this approach uses a pretrained language model — in this case, HyeBERT, an Armenian-language BERT — to convert each document into a fixed-length numerical vector, and then trains a standard classifier on those vectors. BERT processes text as a sequence of tokens and outputs a contextualized representation for each token, so the same word receives a different vector depending on what words surround it. The first token in every input is a special [CLS] token which BERT is trained to use as a summary of the entire sequence. By extracting this [CLS] embedding, I feed a document into BERT, let it process all the tokens in context, and then take the 768-dimensional vector corresponding to that first [CLS] token as my document-level representation (Devlin et al. 2019).

The third approach is *fine-tuned BERT*. In this case, instead of treating BERT as a fixed feature

[1] I constructed this the stopword list manually, as there is currently no extensive, standardized stopword list publicly available for the Armenian language. The list is available in the replication materials.

extractor, I allow the model to partially retrain itself on my labeled data so it can learn the specific political dialect of Armenian civil society. Specifically, I unfreeze the last two of twelve transformer layers and add a classification layer on top. This allows the model to adjust what it "pays attention to" when reading the texts, shifting from a general understanding of Armenian to a specialized understanding of how government criticism is institutionally framed. I evaluate whether this additional flexibility improves performance in Section 3.

Classifiers. I test five classifiers across two text representations identified above. Since the training data is largely balanced, I do not apply class reweighting, as this would bias the predictions towards the critical class and inflate the criticality estimates for the entire corpus.

The first model, Naive Bayes, is applied to the bag-of-words representation, since it relies on an independence assumption between words that does not hold for dense embeddings. It's called a generative classifier because instead of asking the direct question, "Given these words, is this document critical?", it asks the inverse question: "If the document were critical, what words would I expect to see?" (James et al. 2021). Using Bayes' theorem, the model learns to identify which words are more common in critical documents than in non-critical ones and classifies each new text based on the pattern that most closely matches it.

The remaining four classifiers are applied to the frozen BERT embeddings and largely differ in their flexibility. Logistic Regression draws a linear boundary in the embedding space, while tree-based models are designed to capture nonlinear interactions between dimensions. Logistic regression with L2 regularization serves as a linear baseline model. Gen a document's embedding, it estimates the probability of being critical. To account for nonlinearities, I utilize two tree-based ensembles. Random forest builds 500 independent decision trees on bootstrapped samples, ensuring stable results thanks to the "wisdom of the crowd" logic (Montgomery and Olivella, 2018). In contrast, gradient boosting uses a sequential approach, building 300 trees where each new tree focuses on documents that previous trees misclassified, using a learning rate of 0.05 and 80% subsampling (Arnold et al., 2024).

Finally, a Stacking Ensemble combines all three BERT-based classifiers by feeding their cross-validated predicted probabilities as input features into a Logistic Regression meta-learner. In this way, the ensemble optimally weights each classifier's strengths rather than committing to a single model (van der Laan, Polley, and Hubbard 2007). This is particularly important to test, as Abarna et al. (2022) demonstrate that using a stacking ensemble approach significantly minimizes prediction errors, such as log loss, while consistently improving overall classification accuracy and F1-scores compared to using individual baseline models.

Evaluation. To systematically assess these models, I employ 5-fold stratified cross-validation, repeatedly holding out one-fifth of the data to estimate how well each algorithm generalizes to unseen documents (James et al. 2021). Ultimately, the central comparison in this analysis is not just between the classifiers themselves, but between the two text representations, which I expect to be the primary driver of performance differences. Four standard metrics are used to evaluate the models. Accuracy measures the overall proportion of correctly classified documents. While this metric can sometimes be misleading in highly imbalanced datasets, our training data is relatively balanced (49.2% critical), making accuracy a highly relevant performance measure. Given that 56.5% of the training data contains critical documents, a naive model that simply labeled every

document as "critical" would automatically achieve 56.5% accuracy without learning any linguistic patterns. The remaining three metrics focus specifically on the critical class and measure two different types of errors. Precision evaluates the reliability of the model's positive flags. It helps us answer the following question: "What proportion of all documents labeled as critical are actually critical?" High precision is necessary, as too many false positives would artificially inflate the overall estimate of government criticism across the corpus. Recall measures a different concern: "What proportion of all critical documents that exist are successfully detected by the model?" High recall ensures that the model does not miss genuine disagreement, as too many missed instances would lead to a persistent underestimation of government criticism. Finally, the F1-score calculates the harmonic mean of precision and recall, providing a single summary statistic that effectively balances the tradeoff between false positives and missed cases.

$$\text{Accuracy} = \frac{TP + TN}{\text{Total}} \qquad \text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN} \qquad F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

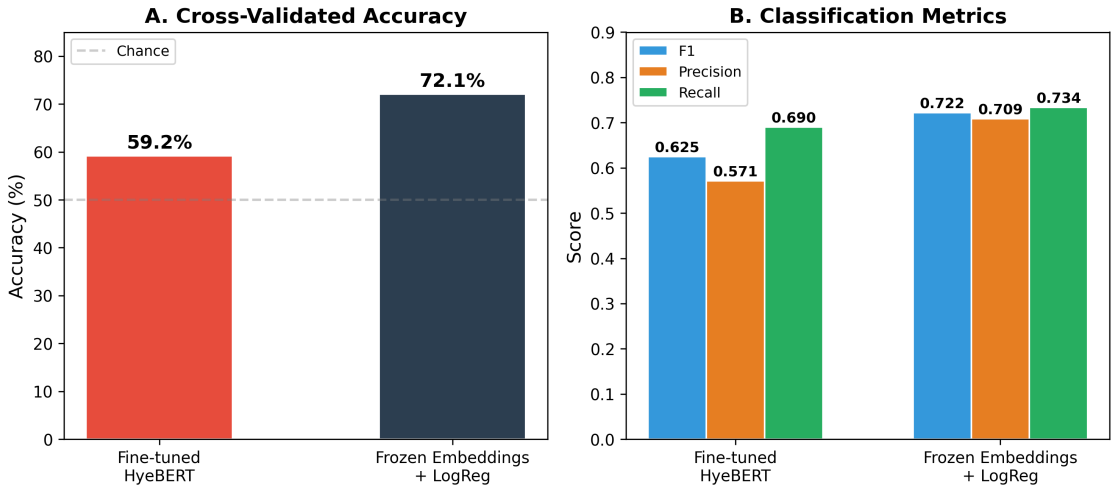
3 Results

3.1 Classification Performance

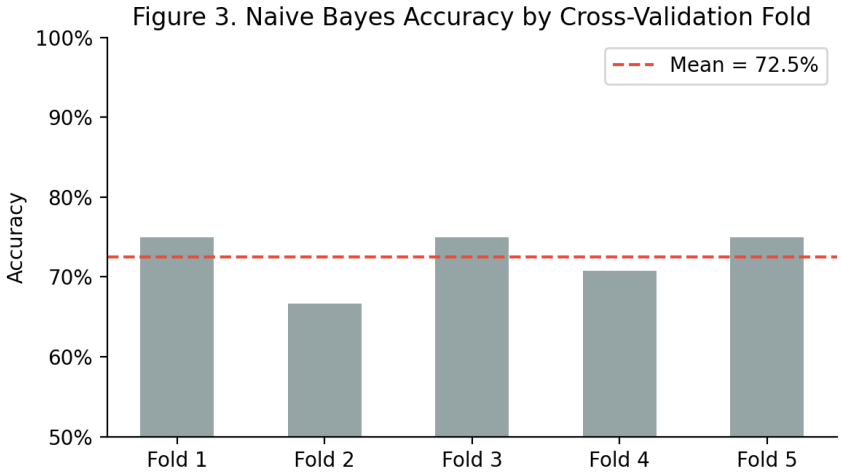
First and foremost, the results indicate that while computationally intensive, the fine-tuning of the BERT architecture is inferior to extracting the frozen embeddings approach. As shown in Table 3, fine-tuning perfoms poorly, with an accuracy of 59.2%, and an F1-score of 0.625. This is considerably worse than simply pairing frozen embeddings with a Logistic Regression baseline that has a 72.1% accuracy, F1 of 0.722. The reason for this failure is simple and obvious. Our hand-coded training dataset contains only 642 examples, so the immense parametric flexibility of fine-tuning leads to model overfitting. As a result, instead of internalizing generalizable linguistic patterns, the model captures the noise. Consequently, in all subsequent stages of the BERT-based analysis, I rely exclusively on frozen embeddings (Figure 9).

Table 3. Fine-tuning vs. feature extraction (5-fold CV)

Method	Accuracy	F1	Precision	Recall
Fine-tuned HyeBERT (last 2 layers)	59.2%	0.625	0.571	0.690
Frozen embeddings + Logistic Regression	**72.1%**	**0.722**	**0.709**	**0.734**



To establish a baseline, I apply Naive Bayes to the bag-of-words representation. As shown in Figure 3, Naive Bayes achieves a reasonable cross-validated accuracy of 72% to 74% . However, one should note that such a sparse representation is limited by a very critical analytical blind spot. In practice, it struggles to distinguish the target of criticism. Because documents criticizing Russia and documents criticizing the Armenian government use much of the same vocabulary, a word-level model cannot reliably separate them. That is precisely why BERT is capable of capturing rich contextual meaning and, as a result, demonstrates improved accuracy.



Having established the superiority of the contextual representation, I test more flexible classifiers on the frozen embeddings. Random Forest achieves 77.6% accuracy and an F1 of 0.811, the highest among the individual methods. Gradient Boosting falls slightly behind at 76.7% accuracy. Lastly, the Stacking Ensemble matches Random Forest at 77.6% accuracy but exhibits the lowest variance across folds. This essetnally indicates that, within the set of classifiers considered and given the size of the training data, no single classifier holds a hidden, untapped advantage. Ultimately, all four BERT-based methods perform within a remarkably narrow band. As demonstrated by Wang (2025), this "ceiling effect" constitutes a theoretical characteristic of self-supervised data representation. Since architectures such as BERT perform the bulk of their computational work during the

pre-training phase, they project text into a representation space that is already highly informative, leaving relatively little room for downstream classifiers to differentiate themselves.

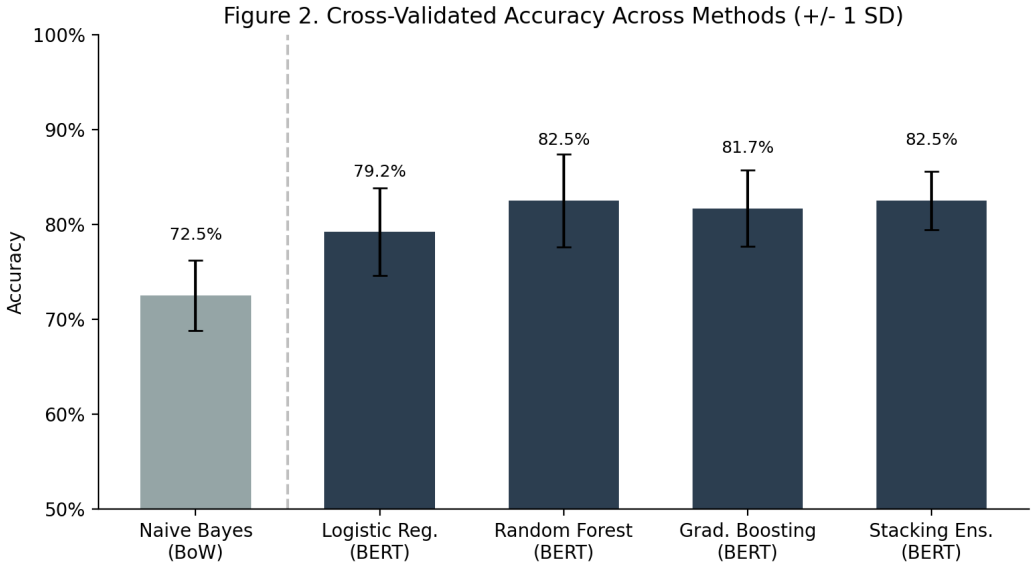
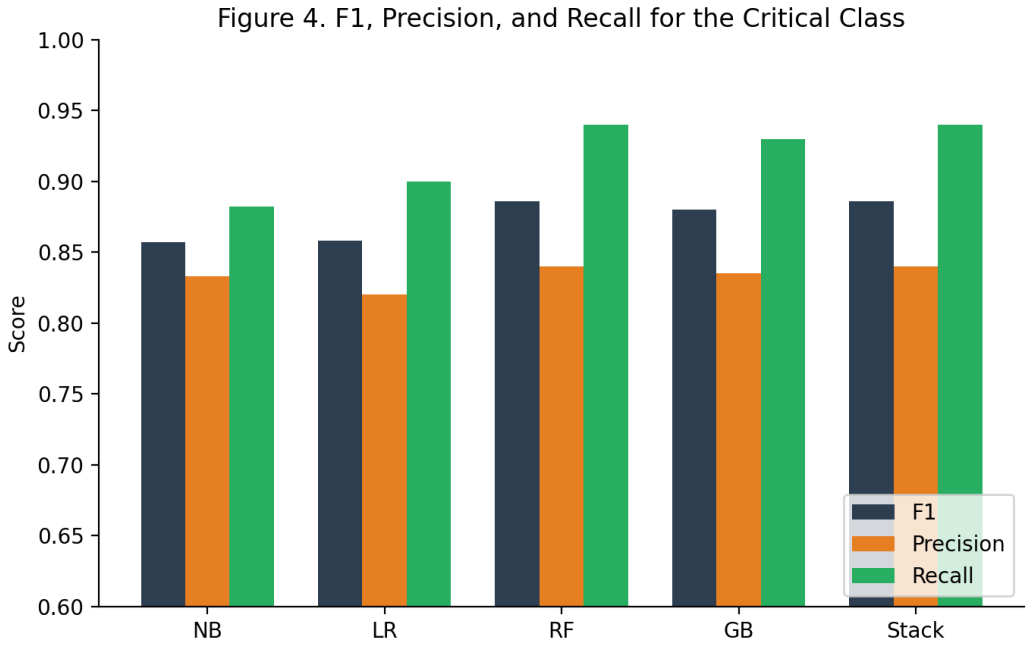
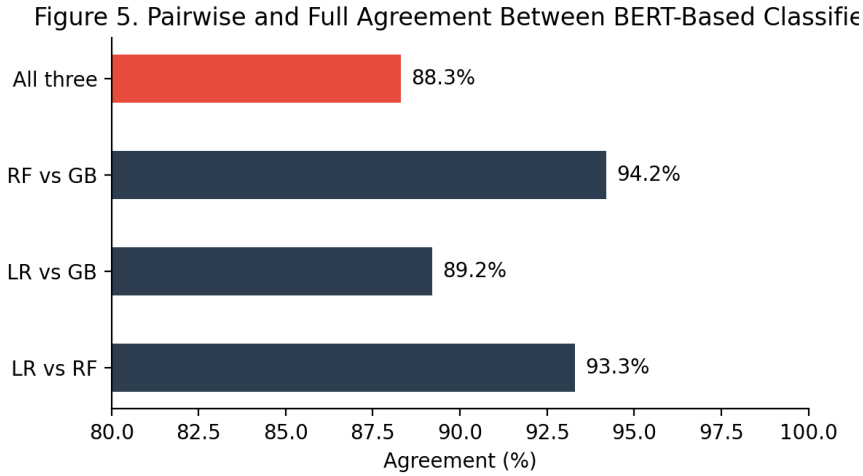


Table 1. Classification performance across methods

Method	Representation	Accuracy	F1 (Critical)
Naive Bayes	Bag-of-words	72.5% (SD 3.7)	0.857
Logistic Regression	BERT embeddings	76.4% (SD 3.4)	0.799
Random Forest	BERT embeddings	77.6% (SD 3.4)	0.811
Gradient Boosting	BERT embeddings	76.7% (SD 3.0)	0.805
Stacking Ensemble	BERT embeddings	77.6% (SD 2.5)	0.811

Although Naive Bayes achieves a higher F1 score for the critical class (0.857), this reflects the model's tendency to systematically over-predict criticism due to its inability to distinguish specific targets. When applied to the full corpus for the substantive analysis, this bias would artificially inflate aggregate criticality estimates. Therefore, given the relatively balanced training set, overall accuracy provides a more reliable basis for comparng representation performance in this specific context.





4 Application: Preliminary Analysis

Having established that the classifiers produce reliable predictions and that the choice of algorithm does not meaningfully affect the results, I now turn to the substantive question: has CSO criticism of the Armenian government changed since the 2018 Velvet Revolution? Classifying the full corpus I apply the Random Forest model to classify the full corpus of approximately 4,100 documents. Figure 6 shows the predicted criticality share by CSO-year.To formally test whether criticism changed after the revolution while accounting for variation in document volume, I estimate a document-level logistic regression with organization fixed effects, that is, sepate intercepts for each CSO that absorb baseline differences in criticality.

Figure 6. Predicted Criticality by CSO-Year

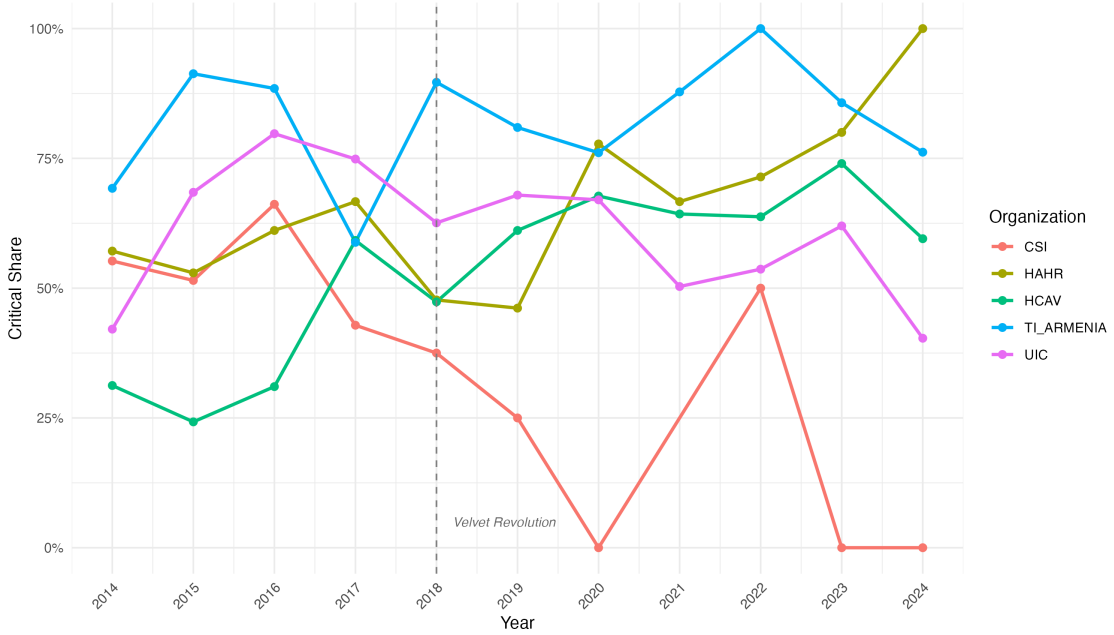


Table 2 disaggregates the results and reveals that each organization followed a very different trajectory after the revolution.

Table 2. Pre- vs. post-2018 changes by organization

Organization	Pre-2018 share	Post-2018 share	Change	Pre-2018 docs	Post-2018 docs	Volume change
CSI	54.4%	17.2%	-37.2 pp	456	29	-94%
HAHR	52.8%	62.7%	+9.9 pp	159	59	-63%
HCAV	42.0%	65.3%	+23.3 pp	305	1,254	+311%
TI Armenia	82.4%	84.4%	+2.0 pp	108	186	+72%
UIC	69.2%	57.6%	-11.6 pp	712	826	+16%

The table above indicates a divergence, with some CSOs becoming more critical, some staying the same, while others are getting more silent. For example, CSI went from publishing over 200 documents per year before 2018 to fewer than six per year after 2020, effectively a near-total cessation of public activity. HCAV moved in the opposite direction, quadrupling its publication volume and increasing its critical share from 42% to 65%. UIC, the largest organization by volume, showed a modest decline in critical share (69% to 58%) while maintaining stable publication output. TI Armenia remained consistentley critical regardless of the period. HAHR published less but maintained or slightly increased its critical intensity.

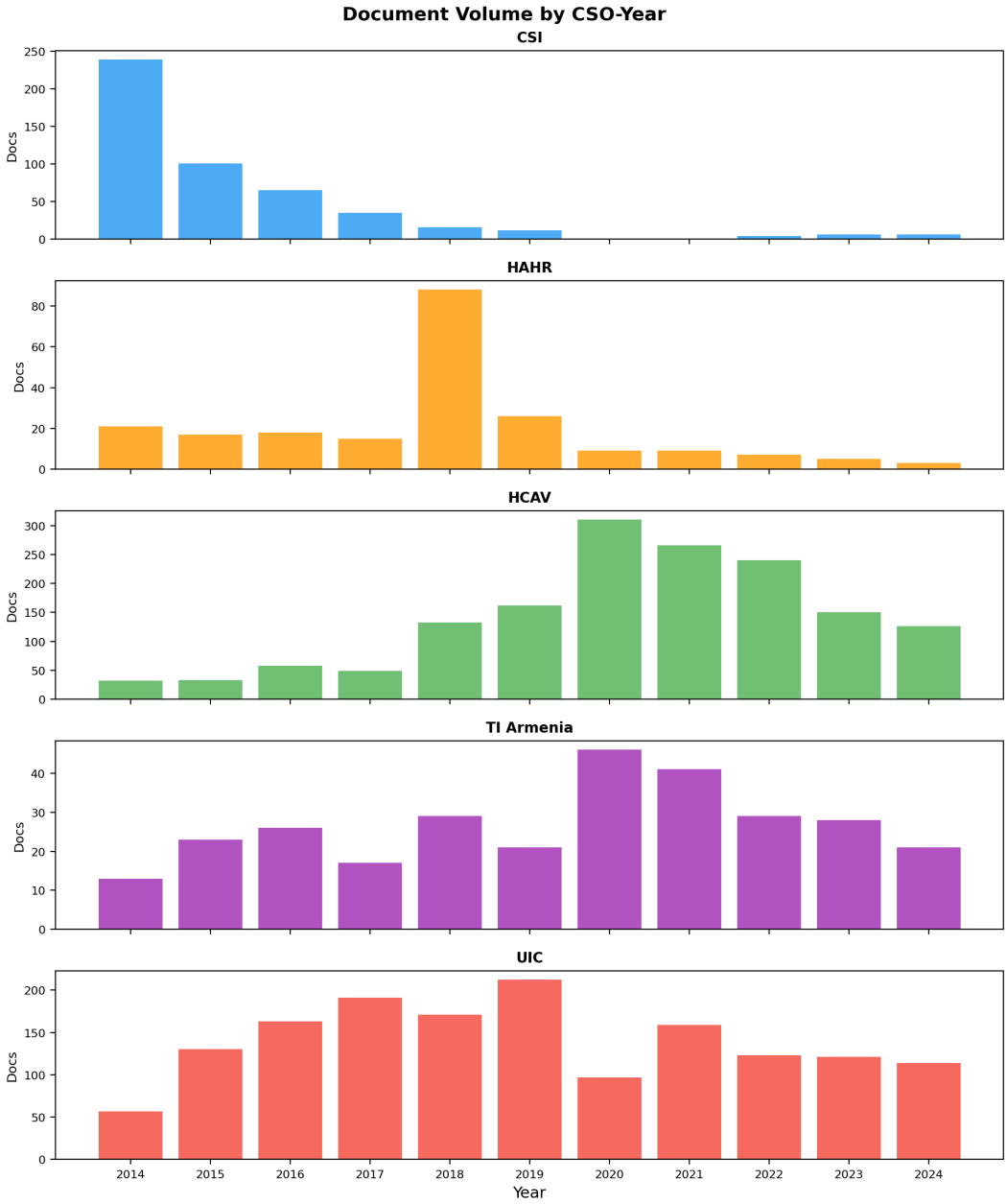
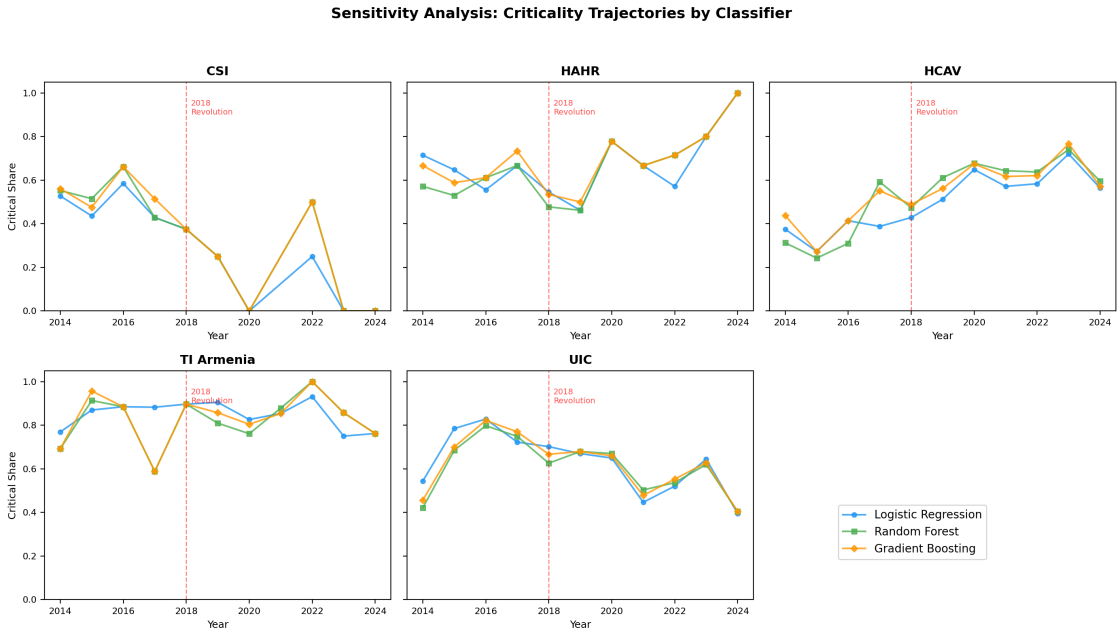


Figure 8 confirms that these organizational patterns are robust to classifier choice. When three BERT-based classifiers are applied independetly to the full corpus, their trajectories overlap closely within each organization, suggesting that the observed heterogeneity is a feature of the underlying data rather than an artifact of any particular algorithm.



5 Conclusion

Having compared classification methods across two text representations to measure government criticism in Armenian watchdong CSO publications, indicates that text representation matters considerably more than algorithmic choice. In practic, transitioning from a sparse bag-of-words approach to dense BERT embeddings gave a 4 percentage point gain in accuracy. Meanwhile, when switching classifiers within that embedding space, we achieve a marginal 1 percentage point improvement. Furthermore, all BERT-based classifiers produced highly consistent criticality trajectories. This further confirm the overall robustness of the measurement strategy.

Building on these methodological findings, a preliminary application of the best-performing classifier to the full corpus of five CSOs revealed that, in the aggregate, the overall volume of CSO criticism remained relatively stable. However, the aggrigation of results masks significant heterogeneity in organizational responses. For instance, while organizations like UIC and CSI decreased their critical coverage, HCAV moved in the opposite direction, both increasing its overall publication volume and becoming more overtly critical. Moving forward, expanding the corpus to include a wider array of civil society organizations will provide the inferential leverage necessary to test competing hypotheses and uncover the specific mechanisms driving these divergent organizational trajectories.

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Supplementary Material

The data, code, and any additional materials required to replicate all analyses in this article are available at https://github.com/Albert-Ananyan/Choosing-a-Classification-Method-for-Measuring-Government-Criticism-in-Armenian-Civil-Society-Texts.